

GIS 5571 Lab 2 Explainer Doc

Analyzing Electric Scooter Availability in Minneapolis

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Lab link:

https://docs.google.com/document/d/18KMk5Vh37cWZJmT_3n-d8i_qXFFhrvfjLtUSMAtxiEc/edit?usp=sharing

Goal:

The goal is to analyze publicly available scooter availability data and compare those patterns to demographic data from the U.S. Census/American Community Survey.

Data:

[U.S. Census Tiger/Line Shapefiles](#); U.S. Census Bureau

- For census tracts of Minneapolis/Hennepin County, MN

[PW Street Centerline](#); City of Minneapolis

- [Alternate Link to data](#)
- For street centerlines of Minneapolis

[City of Minneapolis Scooter Availability \(2024\)](#); City of Minneapolis

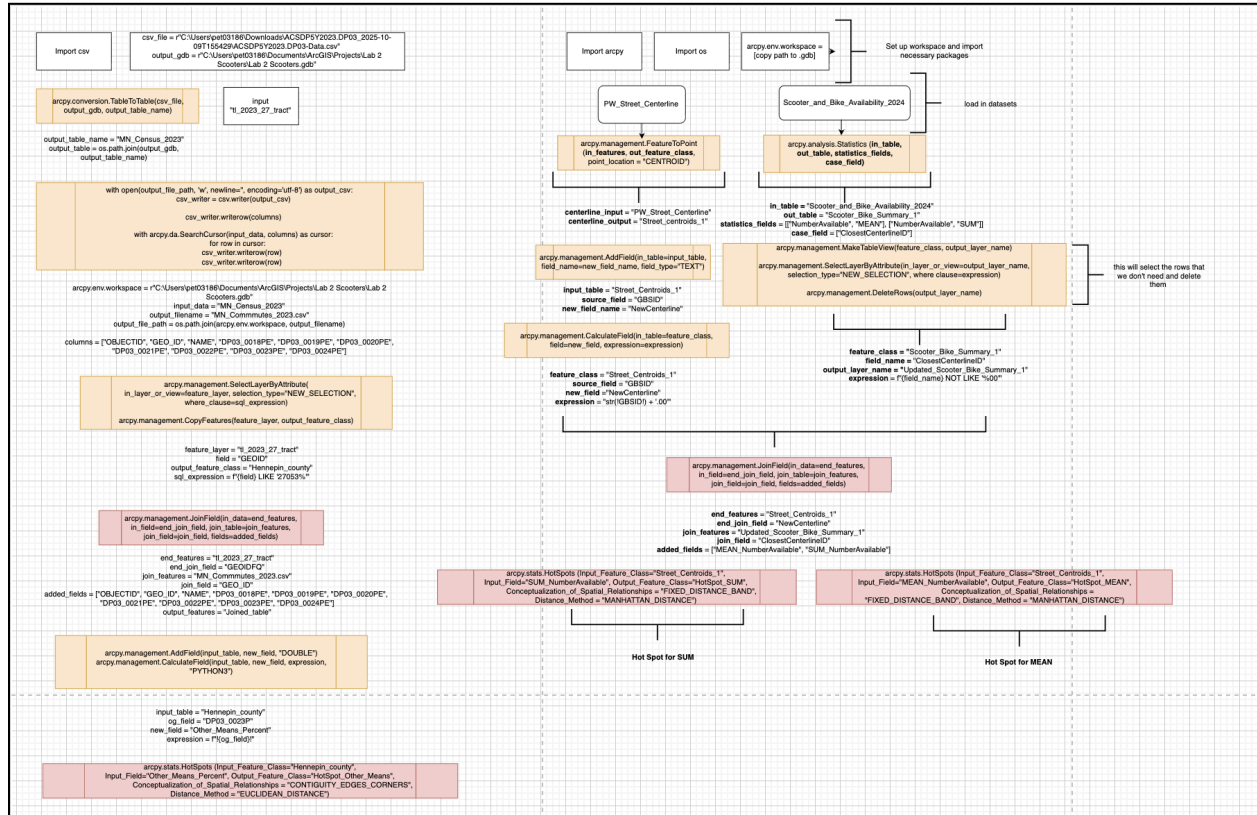
- [Alternate Link to Data](#) (if processing is taking too long)
- Hourly-updated data indicating scooter availability by company and closest street/trail centerline ID
- Scooter locations are aggregated to the nearest street or trail centerline, meaning a join is necessary to spatially enable any scooter data
- In this lab, we will look at availability of scooters (i.e., how many scooters are available at a given location and time), however, another common use would be to look at [individual scooter trips](#) (e.g., each record includes origin/destination locations and start/end timestamps).
- Note: Download the geodatabase file. CSV fields cannot be edited in ArcGIS Pro, Shapefile is too large.

American Community Survey Demographics (2024); U.S. Census Bureau **(for week 2)**

- [2023 ACS 5-year Estimates Data Profiles](#)
 - Justification: most recent 5 year estimate available, so it should have the most up to date commuter information

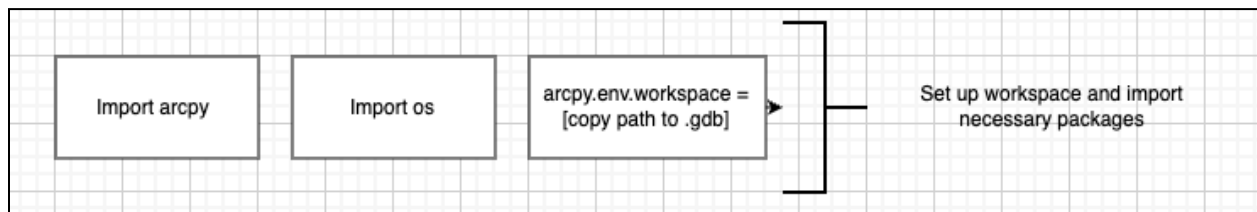
- [TI_2023_27_tract](#)
 - Justification: the 2023 tract data is recent, and also the same published year as our demographic data, preventing any possible joining errors.

Data Flow:

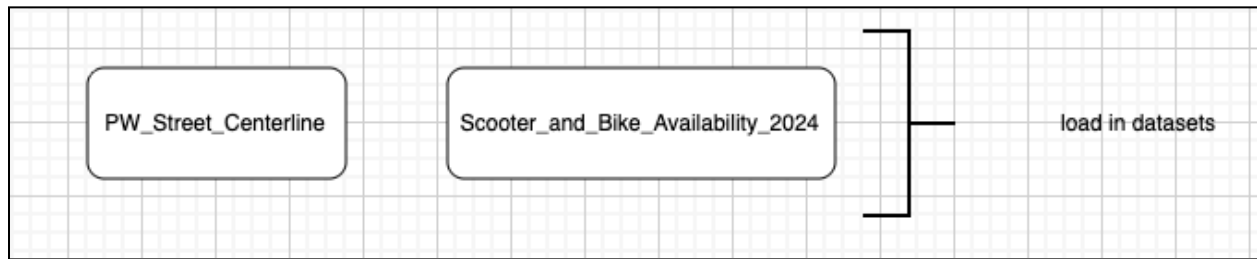


Workflow:

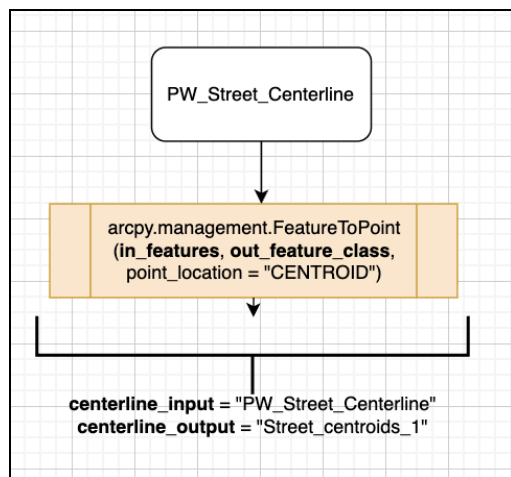
We begin by importing the necessary packages and setting up our workspace, we will by using the arcpy package:



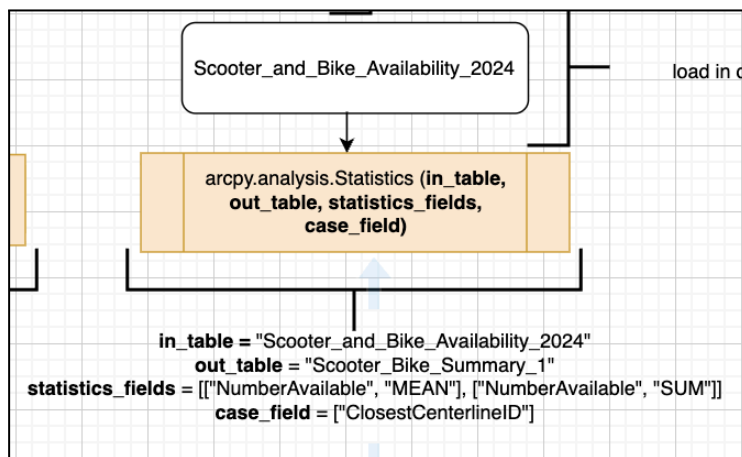
Load in necessary datasets. We will be using Scooter_and_Bike_Availability_2024 from the city of Minneapolis, as well as the PW_Street_Centerline file, which will show us the centerlines of streets throughout the city:



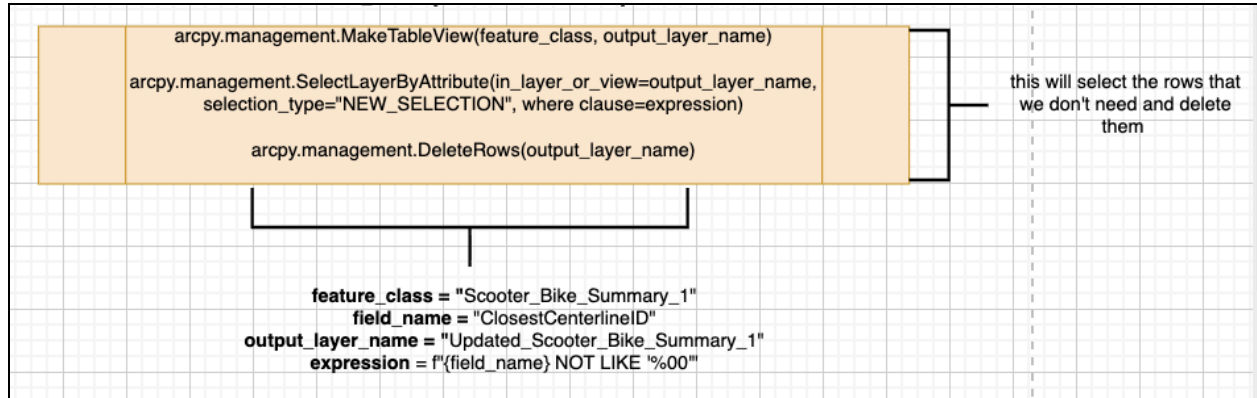
To start, we will use the Feature To Point function in arcpy to convert our centerlines to centroids. Input/output details are specified below:



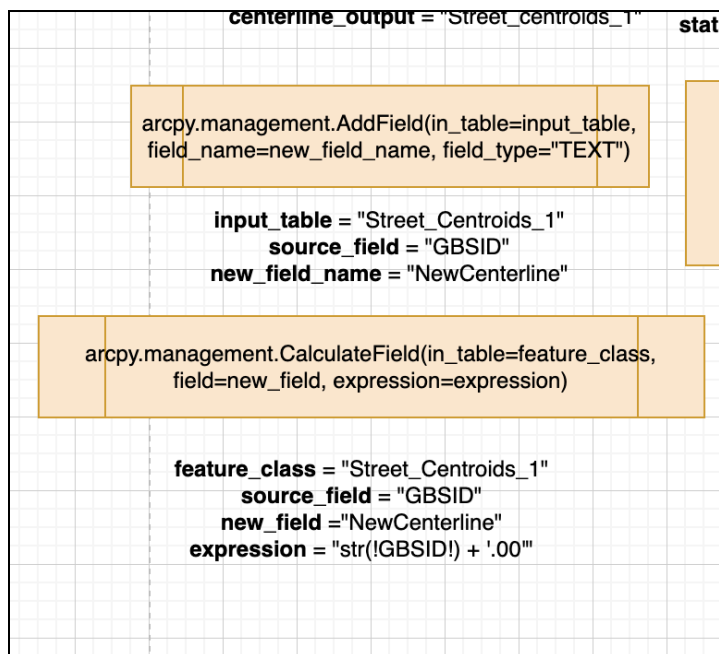
Then, we will start to work with our Scooter_and_Bike_Availability_2024 dataset using the arcpy.analysis.Statistics function. We will calculate the mean and sum values from the "NumberAvailable" field. Parameters are specified below:



In order to join our datasets to create hotspots, we need to first remove the trail centerlines from our dataset. This can be done by combining the three functions below:



In addition, before our final join, we need to add “.00” to the GBSID field in the Street Centroids dataset so that it will match the ID’s in the Updated_Scooter_Bike_Summary_1 file. The two functions below will add a field to the dataset and calculate the new values/populate them into the new field:



Now, we join our datasets together based off their common fields, without forgetting to include the added calculated fields (mean & sum):

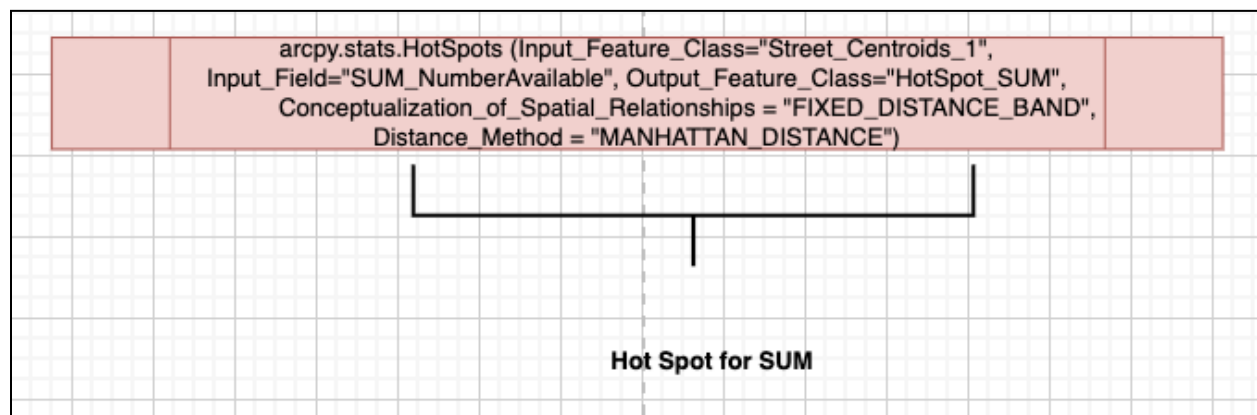
```

    arcpy.management.JoinField(in_data=end_features,
                               in_field=end_join_field, join_table=join_features,
                               join_field=join_field, fields=added_fields)

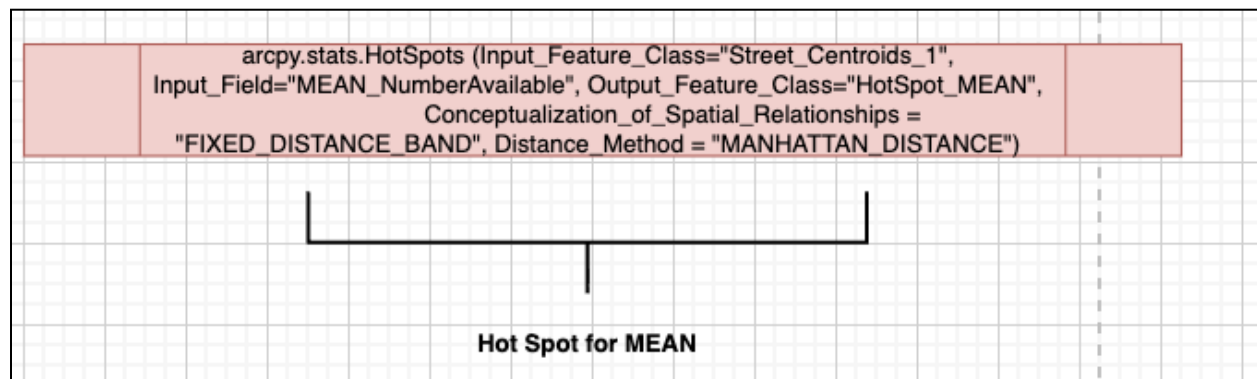
    end_features = "Street_Centroids_1"
    end_join_field = "NewCenterline"
    join_features = "Updated_Scooter_Bike_Summary_1"
    join_field = "ClosestCenterlineID"
    added_fields = ["MEAN_NumberAvailable", "SUM_NumberAvailable"]

```

Hotspots for sum:



Hotspots for mean:

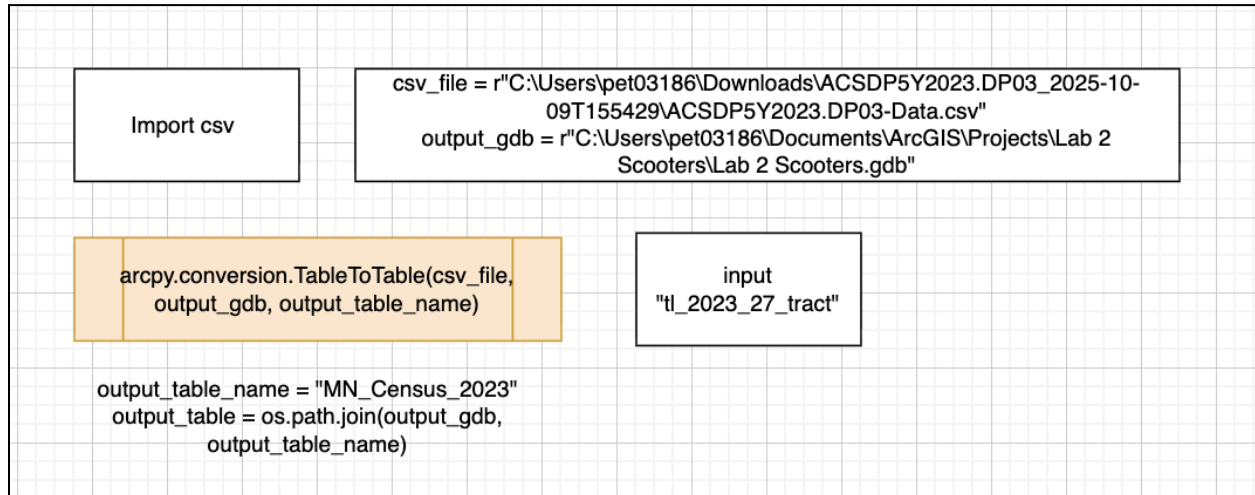


In order to repeat the same process with a different scooter dataset with ease, wrap all of the above steps together in a def (scooter_bike_processing(scooter_input_dataset): function, replace our original dataset with scooter_input_dataset, and call the function using:

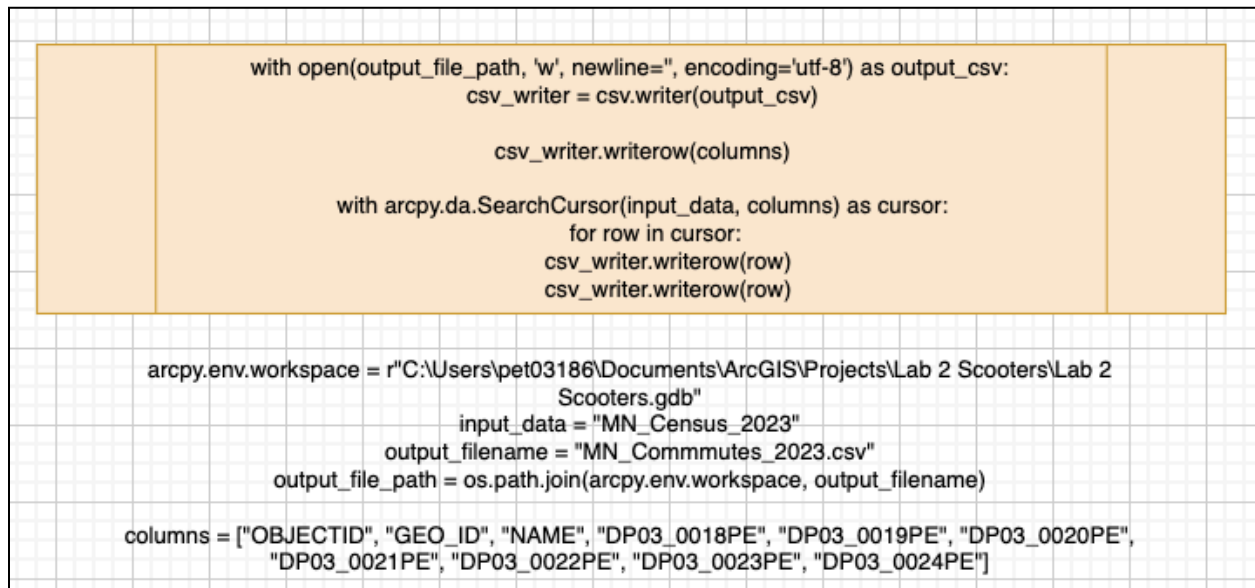
```
scooter_input_dataset = "Scooter_Availability_2020"  
scooter_bike_processing(scooter_input_dataset)
```

ACS DATA

Import necessary packages, including csv, set the path to the csv file (here we are looking at a 5 year summary of data from 2023), and the output_gdb. We will use the TableToTable function to make our csv file editable:



Next, we will create a new csv file (to make analysis easier, there are a lot of fields in the original file) and rewrite the fields in “columns” into the file. Our output file name will be MN_Commmutes_2023.csv



Next, we will filter/clean our data by selecting the census tracts that are within Hennepin County only. These begin with the numbers “27053” in the GEOID field.

```

arcpy.management.SelectLayerByAttribute(
    in_layer_or_view=feature_layer, selection_type="NEW_SELECTION",
    where_clause=sql_expression)

arcpy.management.CopyFeatures(feature_layer, output_feature_class)

```

```

feature_layer = "tl_2023_27_tract"
field = "GEOID"
output_feature_class = "Hennepin_county"
sql_expression = f'{field} LIKE '27053%'

```

Then, we will join our 2023 census tract file with our commuting data so that we can create hotspots.

```

arcpy.management.JoinField(in_data=end_features,
    in_field=end_join_field, join_table=join_features,
    join_field=join_field, fields=added_fields)

```

```

end_features = "tl_2023_27_tract"
end_join_field = "GEOIDFQ"
join_features = "MN_Commmutes_2023.csv"
join_field = "GEO_ID"
added_fields = ["OBJECTID", "GEO_ID", "NAME", "DP03_0018PE", "DP03_0019PE", "DP03_0020PE",
    "DP03_0021PE", "DP03_0022PE", "DP03_0023PE", "DP03_0024PE"]
output_features = "Joined_table"

```

Repeat the step below 4 times, for columns "DP03_0023P", "DP03_0022P", "DP03_0021P", and "DP03_0018P". These represent percent of people that used other means to commute to work, walked, used public transit, and total percent of people that commute to work, respectively.

```
arcpy.management.AddField(input_table, new_field, "DOUBLE")
arcpy.management.CalculateField(input_table, new_field, expression,
                                "PYTHON3")
```

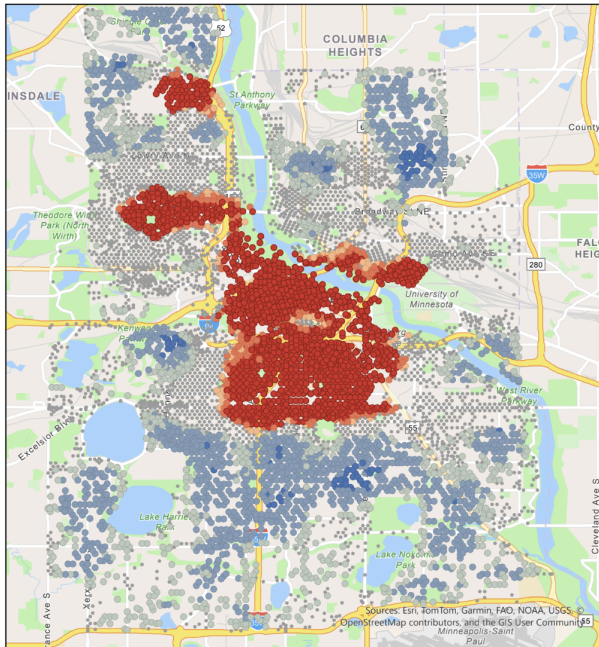
```
input_table = "Hennepin_county"
og_field = "DP03_0023P"
new_field = "Other_Means_Percent"
expression = f"!{og_field}!"
```

Finally, to create our hotspots, we will use the function below 4 times, replacing the input_field with each of the four fields:

```
arcpy.stats.HotSpots (Input_Feature_Class="Hennepin_county",
                      Input_Field="Other_Means_Percent", Output_Feature_Class="HotSpot_Other_Means",
                      Conceptualization_of_Spatial_Relationships = "CONTIGUITY_EDGES_CORNERS",
                      Distance_Method = "EUCLIDEAN_DISTANCE")
```

Result:

Hot Spots for Sum of Scooter and Bike Availability in Minneapolis



HotSpot_SUM

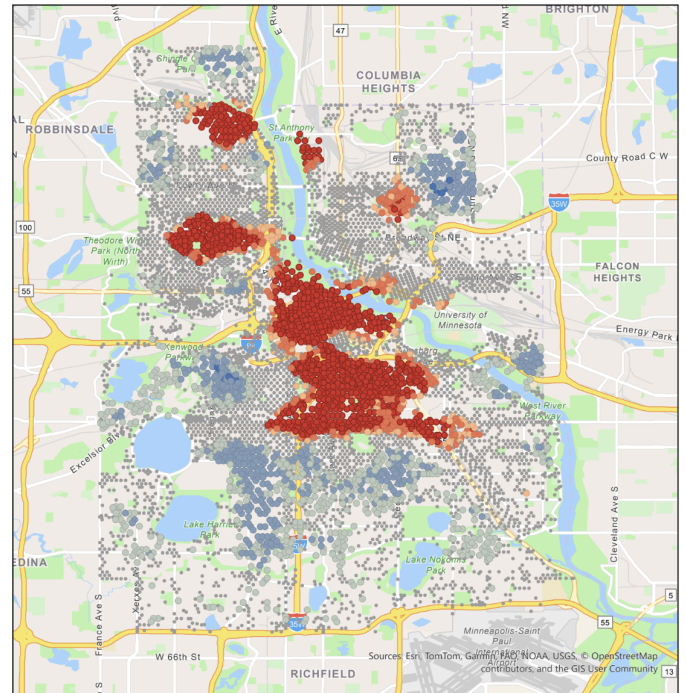
Gi_Bin

- Cold Spot with 99% Confidence

- Cold Spot with 95% Confidence
- Cold Spot with 90% Confidence
- Not Significant

- Hot Spot with 90% Confidence
- Hot Spot with 95% Confidence
- Hot Spot with 99% Confidence

Hot Spots for Mean Scooter and Bike Availability in Minneapolis



HotSpot_MEAN

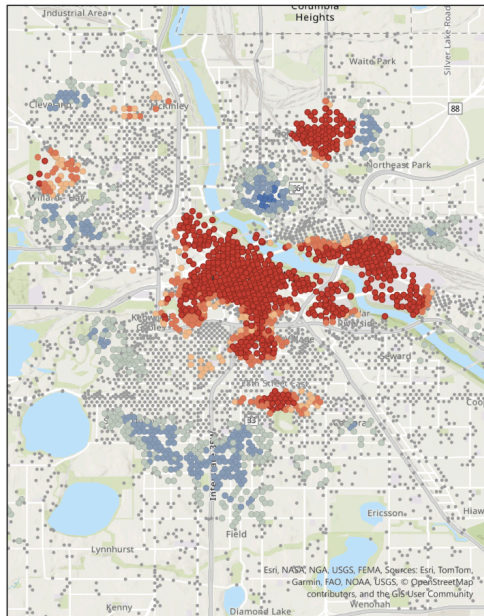
Gi_Bin

- Cold Spot with 99% Confidence

- Cold Spot with 95% Confidence
- Cold Spot with 90% Confidence
- Not Significant

- Hot Spot with 90% Confidence
- Hot Spot with 95% Confidence
- Hot Spot with 99% Confidence

2020 Scooter Bike Hotspots (SUM)

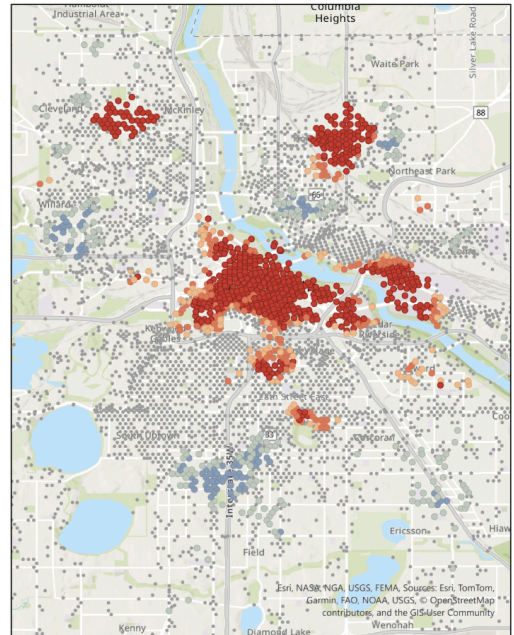


HotSpot_SUM

Gi_Bin

- Cold Spot with 99% Confidence
- Cold Spot with 95% Confidence
- Cold Spot with 90% Confidence
- Not Significant
- Hot Spot with 90% Confidence
- Hot Spot with 95% Confidence
- Hot Spot with 99% Confidence

2020 Scooter Bike Hotspots (MEAN)



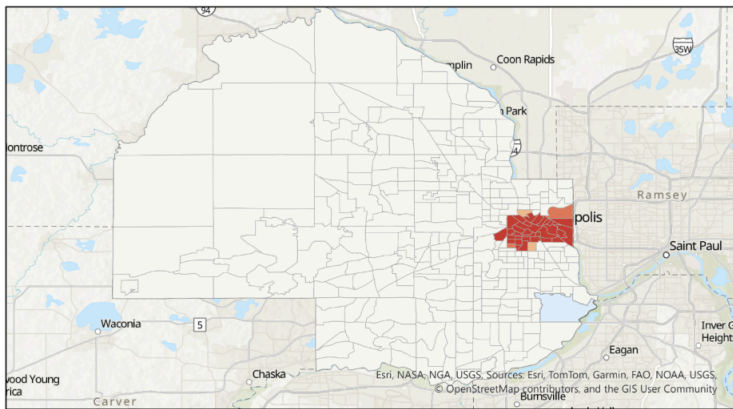
HotSpot_MEAN

Gi_Bin

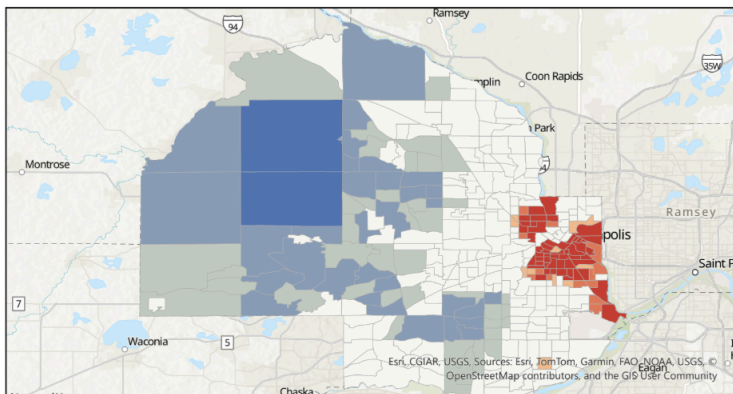
- Cold Spot with 99% Confidence
- Cold Spot with 95% Confidence
- Cold Spot with 90% Confidence
- Not Significant
- Hot Spot with 90% Confidence
- Hot Spot with 95% Confidence
- Hot Spot with 99% Confidence

Hotspot Comparison for Walked to Work/Used Public Transit

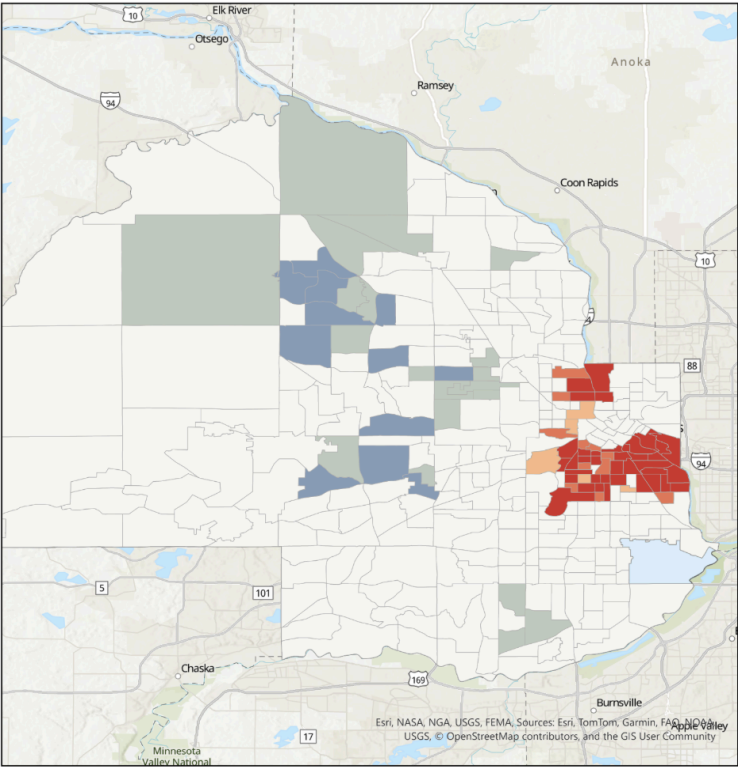
Walked



Public Transit



Hotspots - Other means of transportation



HotSpot_Other_Means
Gi_Bin

Cold Spot with 99%
Confidence

Cold Spot with 95%
Confidence

Cold Spot with 90%
Confidence

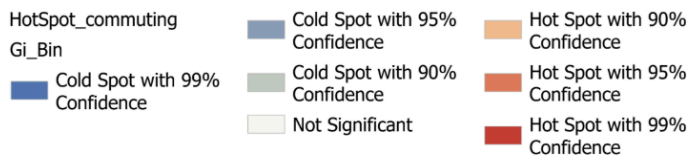
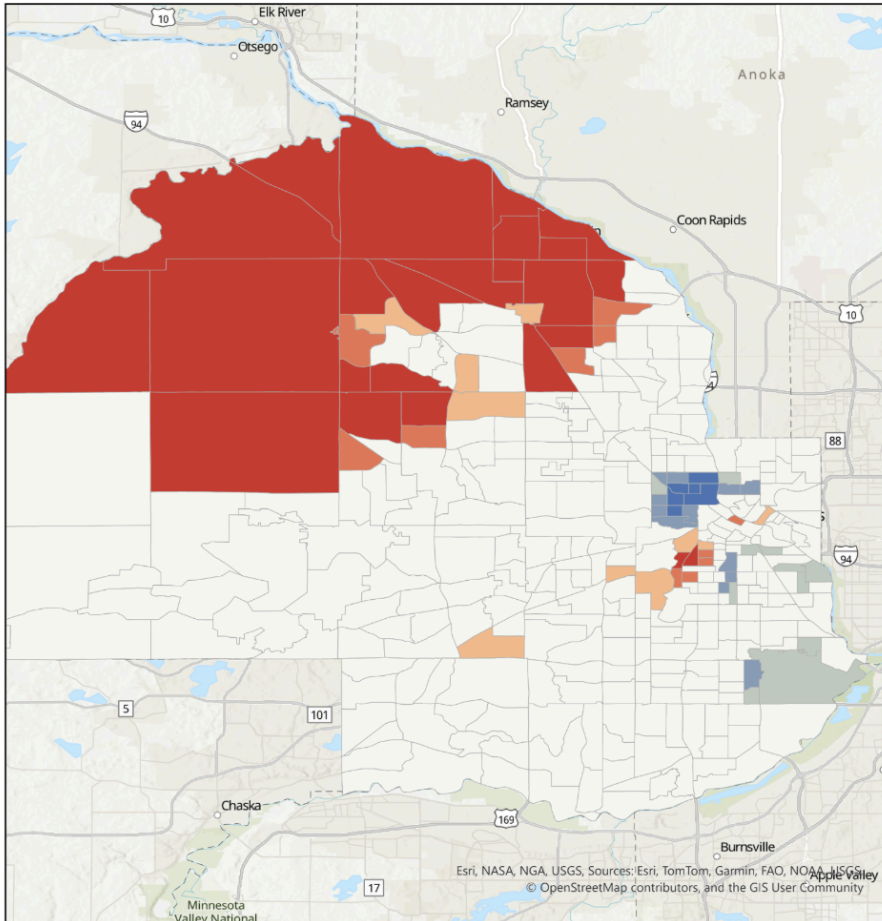
Not Significant

Hot Spot with 90%
Confidence

Hot Spot with 95%
Confidence

Hot Spot with 99%
Confidence

Hotspots - Percent Commuting (Over 16 Yrs. Old)



Results interpretation:

There were interesting changes in the hotspots for mean and sum of scooter availability between 2020 and 2024. I thought that 2020 would be an interesting year to analyze because of the pandemic. Would the hotspots be more concentrated, or more spread apart during quarantine, for instance? The data seems to show that both the mean and sum availability hotspots have spread to areas further outside downtown in recent years, whereas in 2020 they were mostly clustered in the downtown area.

The demographic dataset that I chose unfortunately did not have a “biked” category, and so the field that I thought would most align with the scooter data would be the people using “other means” to get to work. One could also argue that using a scooter could be interpreted as “public transportation” by some people, but since the scooters are not owned by the city, I think that “other means” is the best fit. And, if we compare the hotspots for other means of transportation to the most recent scooter availability data, there does seem to be a correlation. I admittedly may not have used the best method for hotspot analysis of census tracts, and so another contiguity may have been more appropriate.

I unfortunately was having technical difficulties with the ACS website for my fourth analysis, and so rather than downloading an additional dataset, I thought it would be interesting to look at the percent of total people commuting to work. While this hotspot analysis extended past the city of Minneapolis, I found the distribution interesting. Most of the hotspots are located in the NW suburbs of Minneapolis, and cold spots are found just slightly NW of Minneapolis.

Lab Interpretation & Questions:

1. **Why do you think the centerlines are recommended to be converted to points? What are the pros and cons of doing a shape conversion like that? (hint: try the process without converting to points)**
 - a. In order to do a “hotspot” analysis, it is better to have point/centroid data rather than linear features.
2. **Compare and discuss the differences between the two hotspot analyses. Where are they similar? Where are they different?**
 - a. For the sum hotspot, it looks like larger areas are grouped into the hot and cold spots. When you change the analysis to the mean, the hot/cold spots become narrowed down to smaller sections of the city. While the spread is generally pretty similar, it makes sense that the “average” hotspots would be more condensed than the “sum” hotspots.
3. **When may you choose the mean/average for an analysis? When may you choose the sum?**
 - a. I think that doing a mean analysis may be most beneficial when you are trying to gather location information over a larger span of time or space, whereas the analysis for sum may be better for individual events or an understanding of counts.

4. **Write at least one sentence of your own opinion/interpretation of the workflow/results.**
 - a. **When may it be better to use the dataset on actual scooter rides vs general availability?**
 - i. If there was a large event in downtown Minneapolis (ex. Vikings game) it would be interesting to look at actual scooter rides and see if there is an increase in particular areas vs. looking at general availability.
5. **Does the availability of scooters change over the years? Look at both the spatial distribution of the hotspot and the raw counts in the attribute tables to spot any general trends.**
 - a. Yes, it seems that there is more scooter availability in 2024 in comparison to 2020. (expanded on in results interpretation).
6. **Does the availability of scooters correlate strongly with either of the ACS data variables? Is that surprising? Why or why not do you think that is?**
 - a. Based on looking at the hotspots alone, it does seem like “walking” and using “other means” of transportation correlate strongly with scooter availability. This is not that surprising to me since walking/using a scooter are both efficient means of travelling short distances, and “other means” could have been selected as a commute option by survey respondents that scooter/bike to work.
7. **Which hotspot analysis tool did you choose to use? Why? Did you try other hot spot tools? did you explore changing parameters?**
 - a. I used the arcpy “hotspots” tool. I did explore other options for hotspot analysis, but from the research I did, this tool seemed most appropriate. I did try changing different parameters; for example, I used the Queen contiguity for the hotspots that involved the census tracts because they share edges and corners (I am not sure if this was correct, though).
8. **What other methods other than hot spot analysis could be done for scooter availability data?**
 - a. I think that it would be interesting to see the average distance traveled across different tracts. I would assume that in tracts located closer to downtown, that the average distance traveled would be shorter. However, it would be interesting to see if people in Northeast Minneapolis, for example, are traveling to locations within NE, or if they are using scooters to travel into downtown.
9. **Are all the hot spot results similar? Are there any standout points or areas that are unique/surprising in your opinion?**
 - a. While the hotspot results for scooter availability are not very surprising, I did find that the small changes between hotspots in 2020-2024 were interesting. The most interesting hotspot analysis to me was the percent of total commuters; I did not expect it to be so condensed in the NW corner of Hennepin County.
10. **Write at least one sentence of your own opinion/interpretation of workflow/results.**
 - a. I thought this analysis was overall pretty interesting, and I enjoyed working through the process/thinking about other creative methods of analysis I could do after completing the workflow.

11. [optional] Write at least one sentence your thoughts on the lab itself. Structure, style, deliverables, challenges, etc. How long did the lab take you to complete?

- a. The lab took me a pretty long time to complete given all of the components. That being said, I can't complain because at the start of the project I felt very intimidated by the analysis, and working through it myself was very satisfying once I reached the end goal! I think a deeper look into efficient hotspot analysis would be beneficial for me (I read through some of the documentation, but I learn best by looking at scenarios/examples).